

Exact Feasibility of Resistive and AC Power Networks

Abstract

Exact power-flow feasibility can remain difficult under severe graph and coefficient restrictions. We prove that resistive feasibility is complete for the existential theory of the reals, even on connected simple planar bipartite graphs of maximum degree three, with unit conductance and any prescribed fixed lower bound on girth. The model permits independent positive voltage and signed injection intervals, including simultaneous singletons; their endpoints belong to fixed finite rational sets. The main construction realizes bounded addition and inversion while preserving the entire solution set by rational maps in both directions. It transfers to resistive-line AC networks with zero or one-sided reactive injection bounds under explicit angle conventions. For rational cosine limits in $(-1, 1]$, we give a quadratic existential encoding of consistent real bus angles with linearly many variables and predicates; principal line angles alone can admit winding states. The restricted resistive networks realize every compact semialgebraic feasible-set topology. Rational equivalence is possible precisely for compact basic closed sets defined over $\mathbb{Q}$, and unique voltages can have arbitrary algebraic degree. Separately, the ordinary bounded-degree, fixed-data construction yields explicit infeasible networks with doubly exponentially small injection residuals, matching the general separation scale for that class. Short rational certificates under a specified gap promise and quantitative reactive stability delimit what numerical feasibility evidence can certify.

1 Introduction

Power-flow feasibility asks whether the electrical laws and operating limits of a network can hold simultaneously. Even without switching decisions, power is nonlinear in voltage. This leaves two separate questions: how hard is it to decide exact feasibility, and how much does a small numerical residual establish? We answer both for a resistive model with rational conductances, positive voltage intervals, and signed injection intervals, and transfer the exact classification to several precisely specified AC models. The results identify worst-case obstructions to exact certification; they make no claim about the typical difficulty of operating networks.

Our main result is that exact resistive feasibility is complete for $\exists\mathbb{R}$, the class associated with the existential theory of the reals. For each fixed integer r , hardness holds simultaneously on connected simple planar bipartite graphs of maximum degree three and girth at least r , with unit conductance and interval endpoints from fixed finite rational sets depending only on r (Theorem 5.1). Thus neither large numerical coefficients nor dense graphs are required. The class satisfies $\text{NP} \subseteq \exists\mathbb{R} \subseteq \text{PSPACE}$; its relation to exact real-valued certificates and its range of applications are surveyed by Schaefer et al. [20]. A polynomial-size certificate scheme for every feasible instance, with deterministic polynomial-time verification, would therefore imply $\text{NP} = \exists\mathbb{R}$. The separation of these classes is open.

The principal contribution is an electrical realization of bounded arithmetic that preserves the entire solution set, together with transformations that impose all the graph and data restrictions above. To the best of our knowledge, the prior power-flow results discussed below do not provide such a realization or an $\exists\mathbb{R}$ -completeness theorem for this restricted resistive model. The realization

also supports two further conclusions: restricted networks can have arbitrary compact semialgebraic feasible-set topology, and an infeasible network can admit doubly exponentially small injection residuals. The AC results specify which angle conventions preserve these conclusions. They use established cycle and winding ideas; the contribution there is their exact polynomial encoding and the resulting complexity transfers.

1.1 Results, significance, and proof strategy

The results have three connected parts.

1. *Exact complexity and electrical arithmetic.* A bus with prescribed voltage and injection imposes an affine relation on neighboring voltages. Alternating complement paths supply distinct copies of a source variable without increasing the degree beyond three. A small nonlinear gadget then enforces inversion. Every source solution has exactly one network extension, and designated bus voltages recover the source variables (Section 3). A bounded arithmetic crossover gives planarity; free-injection connectors give connectedness; and harmonic subdivisions give unit conductance, bipartiteness, and prescribed girth (Section 5). All these transformations preserve the rational solution-set correspondence. This identifies a source of hardness that survives simultaneous sparsity and coefficient restrictions. The hard graphs may have many cycles: large girth does not mean that they are trees.
2. *AC angle semantics and solution sets.* A local cosine inequality constrains the principal angle of a line but need not admit real bus angles whose ordinary differences satisfy the same limit. A cycle can wind around the origin while every local inequality holds. We encode the additional zero-winding requirement by a quadratic existential formula with $O(|N| + |E|)$ variables, predicates, and monomial occurrences, including unequal voltage magnitudes and every rational cosine in $(-1, 1]$ (Section 4). For resistive lines, an energy identity forces equal real angles when all reactive injections vanish and all line differences have magnitude less than π . This transfers the resistive classification to AC feasibility, including one-sided reactive intervals of positive width. Separate transfers cover reference-fixed bus-angle boxes and strictly positive principal-angle windows that shrink with the number of buses. The latter use polynomial-bit rational cosines, which need not come from a fixed finite set.

The preserved solution sets yield rational equivalence precisely for compact basic closed sets over $\mathbb{Q}$, that is, sets described by a conjunction of polynomial equalities and weak inequalities with rational coefficients (Section 6). Arbitrary compact semialgebraic sets still occur up to semialgebraic homeomorphism. A rational network can also have a unique feasible voltage vector with a designated voltage generating any prescribed real algebraic number field. These results concern topology and exact algebraic coordinates, beyond a count of isolated solutions.

3. *Limits of residual-based certification.* The ordinary electrical reduction satisfies a two-sided residual bound. An explicit infeasible family with $O(k)$ buses, degree at most three, and fixed numerical data admits injection residuals below 2^{-2^k} while respecting every voltage bound exactly (Section 7). A general polynomial-minimum bound gives the matching doubly exponential scale for this fixed-data bounded-degree class. The residual family is not asserted to have all the additional restrictions of the planar construction. A universal residual threshold for exact feasibility can therefore require exponentially many accuracy bits in the number of buses. This is a limitation of certification by residual size alone, not a running-time lower bound for all exact algorithms. At an explicit gap promise, short rational certificates do exist.

Table 1: Exact feasibility classifications for different power-flow models. The restrictions within each row hold simultaneously.

Result	Model and restrictions	Classification
Bienstock and Verma [5]	Lossless AC; fixed voltage magnitudes; unconstrained reactive power	Strongly NP-hard
Lehmann et al. [17]	AC on stars; fixed voltage magnitudes; active and reactive constraints	NP-hard
Theorem 5.1	Resistive; connected, simple, planar, bipartite; degree at most three; any fixed girth bound; unit conductance; fixed finite endpoint sets	$\exists\mathbb{R}$ -complete
Theorem 4.5	Resistive-line AC; variable magnitudes; zero reactive injections; real line differences in $[-\pi/2, \pi/2]$	$\exists\mathbb{R}$ -complete

An energy estimate also bounds angle deviation and active-power error caused by small reactive injections.

1.2 Prior power-flow results and model scope

The resistive equations are the physical direct-current power-flow model studied by Gan and Low [12]. They differ from the linear “DC approximation” of AC power flow. Gan and Low give sufficient conditions for exactness of a second-order cone relaxation of resistive optimal power flow, including conditions involving voltage upper bounds and injection lower bounds. Our classification allows independent voltage and injection intervals, both of which may be singletons at the same bus. In particular, the reduction uses buses with both prescribed voltage and prescribed injection. It does not assume the hypotheses of those relaxation-exactness results. Section 2 states these input choices explicitly; we do not establish hardness after removing the prescribed quantities used by the construction.

A related resistive model fixes source voltages and asks which constant-power demands can be supplied at load buses. Jeeninga et al. [14, Theorems 3.18 and 3.22] characterize exact feasibility and its interior by matrix-inequality alternatives, and establish convexity of the feasible demand set. Their model has no independent operating voltage bounds or injection restrictions at the fixed-voltage sources. Those restrictions distinguish our input model; signed demands alone do not. An exact matrix-inequality characterization also does not, by itself, establish a polynomial-time Turing algorithm for exact feasibility.

For AC networks, Bienstock and Verma [5] prove strong NP-hardness in a lossless model with fixed voltage magnitudes and unconstrained reactive power. Lehmann et al. [17] prove NP-hardness on trees, already using star networks, in a fixed-magnitude model with active and reactive constraints. Our classification identifies the full existential-real complexity of a different subclass: resistive lines and variable magnitudes, with reactive constraints that force equal angles under the stated angle convention. It neither strengthens the tree restriction in their theorem nor proves $\exists\mathbb{R}$ -hardness for the lossless fixed-magnitude model. The model restrictions are summarized in Table 1.

Cycle consistency and winding are established features of power-flow models. Farivar and Low [11, Theorem 2] characterize branch-flow angle recovery by cycle sums that vanish modulo 2π . Delabays et al. [8] relate loop currents to winding numbers, and Jafarpour et al. [13] develop winding cells and uniqueness, modulo a common rotation, within each cell for monotone flow networks on the torus. In our real-angle model, the chosen short line differences must instead sum to zero as real numbers.

This selects the zero-winding cell. We do not claim that criterion, the existence of winding solutions, or uniqueness in the monotone angular regime as new. Our additional step is to express the required branch choices and zero winding in polynomial size with rational coefficients, without taking an inverse trigonometric function as an exact arithmetic primitive. This makes the $\exists\mathbb{R}$ membership statement precise.

The resistive-AC connection with zero reactive injections already appears in Lavaei and Low [16, Appendix B, Case 2]; weighted sine-coupling equilibria also arise in Dörfler et al. [10]. We give a self-contained energy proof under the real-angle hypothesis needed here. On meshed networks, the principal-angle winding examples in Section 4.4 have zero reactive injection and nonconstant phasors. Thus the angle convention is material to the transfer, rather than a choice of coordinates.

1.3 Algebraic and quantitative precedents

The algebraic study of power flow predates this work. Mareček et al. [18] study polynomial formulations, generic complex solution counts, and positive-dimensional solution sets. Our universality results instead realize prescribed compact feasible sets, with operating inequalities, in the restricted resistive family. They do not assert that such behavior is generic. The arithmetic starting problem is due to Abrahamsen et al. [2, 3], and the planar crossover is from Dobbins et al. [9]. For rational universality we use the conjunction-only arithmetic ideas in Abrahamsen and Miltzow [1], with explicit range checks in Appendix A. The broader rational-equivalence statement in Theorem 1 of that preprint includes arbitrary compact semialgebraic sets. Definition 4 of that preprint uses the same rational-coordinate notion as here: the forward and inverse maps must be defined on their respective sets. Proposition 6.3 and the three-quadrant example rule out that broader scope under this definition. We therefore prove the required basic-closed version directly, without its Boolean preprocessing. Semialgebraic triangulation [19] gives the separate topological conclusion. The new application is the electrical realization under all the stated restrictions, not universality of bounded arithmetic or triangulation itself.

Doubly exponential separation is also known for general polynomial systems: Jeronimo et al. [15] give both a polynomial-minimum bound and a repeated-power example showing the need for that scale. Our contribution is an explicit infeasible electrical family using only fixed data and bounded degree, together with a quantitative bound for the actual arithmetic gadgets. The general issues surrounding short rational points and approximate certificates are studied by Bienstock et al. [7]. Our gap-promise certificate follows from a direct Lipschitz and rounding argument for the bounded quadratic network model; we do not present that general approximation principle as new.

Bienstock and Muñoz [4, Theorem 7 and Corollary 8 of the arXiv version] give LP approximation schemes for polynomial network optimization under bounded treewidth and bounded numbers of local variables and constraints, including AC optimal power flow, with scaled feasibility and optimality tolerances. Those guarantees do not decide exact feasibility. Our promise result likewise does not resolve the unpromised approximate-membership question posed for the lossless fixed-magnitude model in Bienstock and Verma [6, Section 1.3]. The quantitative reactive estimate is an explicit consequence of the energy method, included to delimit the numerical use of the exact AC transfer. Appendix B records reproducible exact-arithmetic checks; these support, and do not replace, the proofs.

2 Exact feasibility and the resistive model

In a resistive network, current depends linearly on bus voltages, but power is the product of voltage and current. This distinction makes exact feasibility a quadratic problem even when the network

has no discrete controls. We study its complexity with interval restrictions on both voltage and net power injection. The model concerns direct-current electrical networks; it is distinct from the linear “DC approximation” of an alternating-current network.

2.1 Input model and complexity convention

Let $G = (N, E)$ be a finite simple undirected graph. Each line $\{i, j\} \in E$ has conductance $g_{ij} = g_{ji} \in \mathbb{Q}_{>0}$. Bus i has voltage bounds $0 < \ell_i \leq u_i$ and injection bounds $p_i^- \leq p_i^+$, all rational and finite. Its net injection at voltage vector $v \in \mathbb{R}^N$ is

$$\mathcal{P}_i(v) = v_i \sum_{j:\{i,j\} \in E} g_{ij}(v_i - v_j). \quad (1)$$

An injection is positive for net supply to the network and negative for net withdrawal. An isolated bus has injection zero. The decision problem RPF asks whether

$$\ell_i \leq v_i \leq u_i, \quad p_i^- \leq \mathcal{P}_i(v) \leq p_i^+ \quad (i \in N) \quad (2)$$

has a solution. Singleton intervals are allowed. We impose neither additional line-flow limits nor a requirement that an injection be fixed whenever a voltage is variable. Thus the input permits voltage-pinned buses with fixed injections, variable-voltage buses with fixed injections, and buses with both quantities in intervals. These choices are material to the reduction.

The graph and its rational data are encoded in binary. The existential theory of the reals, denoted ETR, asks whether an existentially quantified Boolean combination of polynomial equalities and inequalities with integer coefficients is true. The class $\exists\mathbb{R}$ consists of decision problems reducible to ETR by polynomial-time many-one reductions. Rational coefficients are equivalent: each atomic constraint can have its denominators cleared using a positive integer of polynomial bit length. We use the Turing model of computation and the standard inclusions

$$\text{NP} \subseteq \exists\mathbb{R} \subseteq \text{PSPACE}. \quad (3)$$

See Abrahamsen and Miltzow [1, Section 1.1] for these conventions and their background. Exact feasibility quantifies over real solutions, not over decimal approximations. An irrational solution alone does not rule out membership in NP: a certificate need not list rational bus voltages.

Proposition 2.1. *The problem RPF belongs to $\exists\mathbb{R}$.*

Proof. Equations (1)–(2) form a conjunction of rational polynomial inequalities of degree at most two in $|N|$ real variables. Writing each incident-line term explicitly uses $O(|N| + |E|)$ terms and polynomially many bits. Clearing denominators gives an ETR instance. $\square$

The model respects the usual dissipation identity:

$$\sum_{i \in N} \mathcal{P}_i(v) = \sum_{\{i,j\} \in E} g_{ij}(v_i - v_j)^2 \geq 0. \quad (4)$$

Indeed, the two terms of a line sum to $g_{ij}(v_i - v_j)^2$. The injection intervals, rather than a separate balance equation with zero total injection, account for these losses. In particular, permitting signed injections does not remove physical power balance.

2.2 The bounded arithmetic source problem

An instance of ETR-INV consists of named variables $x_1, \dots, x_n$, each restricted to $[1/2, 2]$, and a conjunction of equations of the forms

$$x = 1, \quad x + y = z, \quad xy = 1. \quad (5)$$

Variable names in an equation may coincide. The interval restrictions are part of feasibility, not a promise that all real solutions already lie there. Abrahamsen, Adamaszek, and Miltzow prove this problem $\exists\mathbb{R}$ -complete [2, Definition 5 and Theorem 7 of the STOC/arXiv version]. The journal version is Abrahamsen et al. [3]. Replacing each equation $x = 1$ by $xx = 1$ preserves the solution set, since $x > 0$. Hence we use only addition and inversion equations below. Let $S_\Phi \subseteq [1/2, 2]^n$ denote the solution set of the normalized instance Φ , and let m be its number of equations. We retain all named variables, including those occurring in no equation.

3 A bounded-degree reduction with fixed numerical data

Theorem 3.1. *The problem RPF is $\exists\mathbb{R}$ -complete. Hardness holds for simple graphs with maximum degree three, conductances in $\{1, 2\}$, voltage intervals in*

$$\{[1/2, 2], [1, 1], [1, 4]\},$$

and injection intervals in

$$\{[-85, 85], [-1/2, -1/2], [1/2, 1/2], [-5/2, -5/2], [-1, -1]\}.$$

For an instance of ETR-INV with n named variables and m equations, the reduction constructs at most $n + 16m$ buses and $18m$ lines.

The construction uses a voltage-pinned bus to impose a linear relation on its neighbors, and one variable-voltage bus to impose inversion. Copying values along paths keeps the degree bounded even when a variable occurs arbitrarily often. Throughout this section, a *free injection* means the fixed interval $[-85, 85]$, not an unbounded or omitted constraint.

3.1 Pinned buses and the supply of distinct value copies

A *value bus* has voltage interval $[1/2, 2]$ and free injection. A *pinned bus* has voltage 1 and a specified injection p . If its distinct neighbors carry voltages $a_1, \dots, a_d$ with conductances $h_1, \dots, h_d$, its equation is

$$\sum_{r=1}^d h_r(1 - a_r) = p. \quad (6)$$

In particular, a pinned bus C with injection $-1/2$ and two unit-conductance lines enforces $a + b = 5/2$. Write $\bar{x} = 5/2 - x$; the map $x \mapsto \bar{x}$ is an involution of $[1/2, 2]$.

For each variable x begin with a value bus X_x . Successively extending a path

$$X_x - C_1 - \bar{X}_x - C_2 - X'_x - \dots \quad (7)$$

supplies alternating voltages $x, \bar{x}, x, \dots$. Each C_r has voltage 1, injection $-1/2$, and exactly its two path lines. Every value bus may receive at most one further line from an equation gadget. Here is a deterministic allocation rule. Maintain the current end of each path, its parity (plain or

complemented), and whether its one gadget slot is unused. On a request for a copy, extend the path until the end has the requested parity and an unused slot; allocate that slot and continue from this end for the next request. Each extension adds one pinned bus, one value bus, and two lines. At most two extensions are needed per request. All requested copies are distinct, even if they represent the same variable in the same equation. Unused named variables retain their single isolated bus X_x .

3.2 Addition and inversion gadgets

For $x + y = z$, introduce a pinned bus A with injection $1/2$ and three unit-conductance lines to fresh requested copies of x , y , and $\bar{z}$. Its equation is

$$(1 - x) + (1 - y) + (1 - \bar{z}) = 1/2 \iff x + y = z.$$

For $x + x = z$ the two plain copies of x are distinct buses. The same allocation rule covers $x + y = x$ and $x + x = x$; such equations are correctly infeasible in the positive source interval.

For $xy = 1$, introduce buses I, W, C_I, D with the data in Table 2. Request three distinct complemented value buses: two for x and one for y . Connect them as in Figure 1. If $x = y$, all three still have distinct bus identities.

Bus	Voltage interval	Injection interval	Neighbors (conductance)
I	$[1/2, 2]$	$[-1, -1]$	W (1), C_I (1)
W	$[1, 4]$	$[-85, 85]$	I (1), D (1)
C_I	$[1, 1]$	$[-1/2, -1/2]$	I (1), first $\bar{x}$ copy (1)
D	$[1, 1]$	$[-5/2, -5/2]$	W (1), second $\bar{x}$ copy (2), $\bar{y}$ copy (1)

Table 2: The four new buses for an inversion equation $xy = 1$.

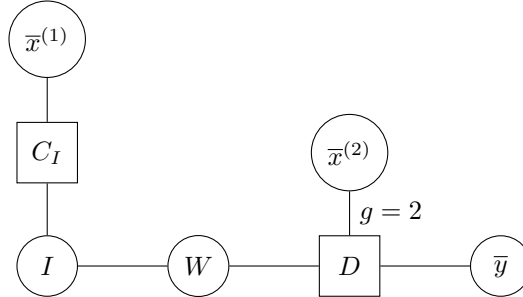


Figure 1: Inversion gadget. Squares have voltage 1; the three upper or rightmost value copies belong to variable paths, whose lines are omitted. Every unmarked line has conductance 1. Superscripts distinguish buses, not numerical values.

The equation at C_I gives

$$(1 - v_I) + (1 - \bar{x}) = -1/2, \quad \text{hence } v_I = x.$$

The injection equation at I is therefore

$$x[(x - v_W) + (x - 1)] = -1, \quad v_W = 2x - 1 + 1/x. \quad (8)$$

Division is valid because $x \geq 1/2$. At D the equation reads

$$(1 - v_W) + 2(1 - \bar{x}) + (1 - \bar{y}) = -5/2 \iff y = v_W - 2x + 1. \quad (9)$$

Together these equations imply $y = 1/x$.

Conversely, if $x, y \in [1/2, 2]$ and $xy = 1$, set $v_I = x$ and $v_W = h(x) := 2x - 1 + 1/x$. Both pinned-bus equations and the injection equation at I then hold. The voltage bound on W is valid: on $[1/2, 2]$, $h'(x) = 2 - x^{-2}$ and $h''(x) = 2x^{-3} > 0$, so its minimum is $2\sqrt{2} - 1$ at $1/\sqrt{2}$ and its maximum is the larger endpoint value, $h(2) = 7/2$. Thus

$$1 < 2\sqrt{2} - 1 \leq v_W \leq 7/2 < 4. \quad (10)$$

3.3 The complete network and proof of equivalence

Construct all variable paths and equation gadgets with the allocation rule above. Path value buses have at most two path lines and one gadget line. Path pinned buses and C_I have degree two; addition buses and D have degree three; I and W have degree two. Distinct requested copies and fresh gadget buses prevent loops and parallel edges. Consequently G is simple and its maximum degree is three.

Every bus voltage belongs to $[1/2, 4]$ and every conductance is at most 2. For every voltage vector in the prescribed box, not just the intended solutions, a free-injection bus satisfies

$$|\mathcal{P}_i(v)| \leq v_i \sum_{j: \{i,j\} \in E} g_{ij} |v_i - v_j| \leq 4 \cdot 3 \cdot 2 \cdot \frac{7}{2} = 84 < 85. \quad (11)$$

The free-injection intervals therefore add no restriction inside that box.

Lemma 3.2. *The constructed network is feasible if and only if $S_\Phi \neq \emptyset$. More precisely, restriction to $(v_{X_x})_{x=x_1, \dots, x_n}$ is a bijection from its feasible voltage set onto S_Φ .*

Proof. Given a feasible voltage vector, put $x = v_{X_x}$ for each named variable. Every path pinned bus has exactly its two prescribed neighbors, so its equation forces successive values to be complementary. Every requested copy thus has its specified value. The addition equation yields $x + y = z$; equations (8)–(9) yield $xy = 1$. This recovers a point of S_Φ .

Conversely, given a point of S_Φ , assign alternating values x and $5/2 - x$ along each variable path, voltage 1 to every pinned bus, and $v_I = x$, $v_W = 2x - 1 + 1/x$ in each inversion gadget. The preceding algebra verifies every fixed-injection equation. The voltage bounds follow from the source interval and (10); the remaining injection bounds follow from (11). An isolated X_x has injection zero and retains exactly the unconstrained source coordinate $x \in [1/2, 2]$.

Finally all bus voltages have been uniquely determined by the recovered source coordinates: path values alternate, pinned values are fixed, and (8) determines W . These two maps are inverse bijections. $\square$

Proof of Theorem 3.1. Each equation requests exactly three value copies. With at most two path extensions per request, the paths contribute at most $12m$ new buses and $12m$ lines in addition to the n initial buses. An addition adds one further bus and three lines; an inversion adds four further buses and six lines. Hence the stated bounds $n + 16m$ and $18m$ hold, including unused variables. All numerical data are exactly those displayed in the theorem. The number of objects is linear in $n + m$; encoding their indices uses $O((n + m) \log(n + m + 2))$ bits, and the construction is polynomial-time. Lemma 3.2 and $\exists\mathbb{R}$ -completeness of ETR-INV give hardness. Proposition 2.1 gives membership. $\square$

Proposition 3.3 (Preservation of the solution set). *The extension map from source solutions to feasible network voltages in Lemma 3.2 is a homeomorphism whose forward and inverse coordinate functions are rational with rational coefficients. Its inverse is restriction to the buses X_x , a coordinate projection.*

Proof. The extension map has coordinate functions x , $5/2 - x$, 1 , and $2x - 1 + 1/x$. They are rational and continuous because $x \geq 1/2$. The inverse projection is linear and continuous. The statement also holds for empty solution sets and for unused source coordinates. $\square$

Corollary 3.4. *If RPF belongs to NP, then $\text{NP} = \exists\mathbb{R}$. In particular, unless these classes coincide, not every feasible rational instance has a polynomial-size certificate verifiable in deterministic polynomial time. The problem belongs to PSPACE.*

Proof. By completeness every problem in $\exists\mathbb{R}$ reduces to RPF. Membership of RPF in NP would imply $\exists\mathbb{R} \subseteq \text{NP}$, and (3) gives equality. The certificate formulation is the definition of NP; the space upper bound follows from the same inclusions. $\square$

The fixed-data conclusion is stronger than a claim that large numerical coefficients can be encoded economically: all numerical choices in the hard instances come from finite sets independent of instance size. Their combinatorial incidence pattern carries the growing input. The theorem uses general graphs and signed injection intervals; it makes no assertion here about a load-only model or a tree restriction.

4 AC feasibility and the meaning of angle limits

The resistive result transfers to an AC model with zero reactive injections. The transfer requires a consistent real angle at every bus: small principal angle differences on individual lines need not fit together around a cycle. We first encode this consistency without transcendental constraints. The encoding works for every line limit strictly below π , including limits larger than $\pi/2$.

4.1 The AC input model

Write $i^2 = -1$. On a simple undirected graph, line $\{i, j\}$ has series admittance $y_{ij} = g_{ij} + ib_{ij}$, where $g_{ij} \in \mathbb{Q}_{\geq 0}$, $b_{ij} \in \mathbb{Q}$, and $y_{ij} \neq 0$. Bus i may have a shunt admittance $s_i = h_i + it_i$, with $h_i \in \mathbb{Q}_{\geq 0}$ and $t_i \in \mathbb{Q}$; absence of a shunt means $h_i = t_i = 0$. These are reciprocal series lines without transformers or phase shifters. A line-charging shunt at a line endpoint may be included in that bus's shunt sum. Let $U_i = v_i \exp(i\theta_i)$, with $v_i > 0$ and $\theta_i \in \mathbb{R}$. With injection positive into the network, the equations are

$$I_i = s_i U_i + \sum_{j: \{i,j\} \in E} y_{ij} (U_i - U_j), \quad P_i + iQ_i = U_i \bar{I}_i. \quad (12)$$

The input gives positive rational magnitude bounds $\ell_i \leq v_i \leq u_i$ and finite rational intervals $[p_i^-, p_i^+]$, $[q_i^-, q_i^+]$ for P_i, Q_i . Singleton intervals are allowed independently for these three quantities. Every line also has a rational number $c_{ij} \in (-1, 1]$, representing $\gamma_{ij} = \arccos c_{ij} \in [0, \pi)$, and the constraint

$$|\theta_i - \theta_j| \leq \gamma_{ij}. \quad (13)$$

The problem ACPF asks for $v \in \mathbb{R}_{>0}^N$ and $\theta \in \mathbb{R}^N$ satisfying these equations and bounds. The rational input contains c_{ij} , not an approximation to γ_{ij} . Angles may be shifted by a common

constant on each connected component; an optional reference angle zero per component only fixes this freedom.

For later use, write $U_i = e_i + if_i$ and introduce a positive magnitude variable r_i . Define

$$H_{ij} = e_i e_j + f_i f_j, \quad D_{ij} = f_i e_j - e_i f_j. \quad (14)$$

Thus $D_{ij} = \text{Im}(U_i \overline{U_j}) = -D_{ji}$. Expanding (12) gives the polynomial expressions

$$P_i = h_i r_i^2 + \sum_j [g_{ij}(r_i^2 - H_{ij}) - b_{ij} D_{ij}], \quad (15)$$

$$Q_i = -t_i r_i^2 + \sum_j [-b_{ij}(r_i^2 - H_{ij}) - g_{ij} D_{ij}], \quad (16)$$

$$r_i^2 = e_i^2 + f_i^2, \quad \ell_i \leq r_i \leq u_i. \quad (17)$$

Here and below, line sums run over neighbors of i . All these constraints have rational coefficients and degree at most two.

4.2 A polynomial encoding of real angle lifts

For two nonzero phasors with non-antipodal directions, let $\delta_{ij} \in (-\pi, \pi)$ be the principal argument of $U_i \overline{U_j}$. The principal-angle limit $|\delta_{ij}| \leq \gamma_{ij}$ is exactly

$$H_{ij} \geq c_{ij} r_i r_j. \quad (18)$$

Indeed, $H_{ij}/(r_i r_j) = \cos \delta_{ij}$, and cosine decreases on $[0, \pi]$. Since $c_{ij} > -1$, antipodal directions are excluded. The positive variables r_i make squaring unnecessary, so the same test covers negative cosines. Condition (18) alone does not encode (13).

Lemma 4.1 (Short-arc crossing count). *Let $U_i, U_j \neq 0$ have non-antipodal directions, and traverse the short oriented arc from $U_j/|U_j|$ to $U_i/|U_i|$. Its signed crossing count of the positive real ray, counting arrival from the open lower half-plane as +1 and departure into that half-plane as -1, is*

$$k_{ij} = \begin{cases} 1, & f_j < 0 \leq f_i \text{ and } D_{ij} > 0, \\ -1, & f_i < 0 \leq f_j \text{ and } D_{ij} < 0, \\ 0, & \text{otherwise.} \end{cases} \quad (19)$$

In particular $k_{ji} = -k_{ij}$. If $\alpha_i, \alpha_j \in [0, 2\pi)$ are the arguments of the endpoints, then

$$\delta_{ij} = \alpha_i - \alpha_j + 2\pi k_{ij}. \quad (20)$$

Proof. Put $a = \alpha_i - \alpha_j \in (-2\pi, 2\pi)$. If $a < -\pi$, then $\delta_{ij} = a + 2\pi \in (0, \pi)$. Necessarily $\alpha_j > \pi$ and $\alpha_i < \pi$, so $f_j < 0 \leq f_i$, and $D_{ij} = r_i r_j \sin \delta_{ij} > 0$. Conversely, $f_j < 0 \leq f_i$ implies $\alpha_j \in (\pi, 2\pi)$ and $\alpha_i \in [0, \pi]$, hence $a < 0$. Within $(-2\pi, 0)$ the additional condition $\sin a > 0$ holds exactly when $a < -\pi$. Thus the first case of (19) is precisely $a < -\pi$. Reversing i, j proves that its second case is precisely $a > \pi$. The remaining case has $|a| < \pi$, including $a = 0$, and needs no wrap. The equalities $a = \pm\pi$ are excluded by the non-antipodal hypothesis. This proves (20) and reversal symmetry. It also checks the axis convention: argument zero belongs to the closed upper half-plane; argument π does too, but the determinant sign prevents counting a crossing of the negative real ray. $\square$

Fix one orientation $j \rightarrow i$ for every line. For any directed traversal of a cycle C , write $\varepsilon_{C,ij} = 1$ if it traverses that line in its chosen orientation and -1 if it traverses it in reverse. Telescoping the α terms in (20) gives

$$\sum_{\{i,j\} \in C} \varepsilon_{C,ij} \delta_{ij} = 2\pi \sum_{\{i,j\} \in C} \varepsilon_{C,ij} k_{ij}. \quad (21)$$

The integer on the right after division by 2π is the winding number. Cycle-angle recovery and winding representations are established in Farivar and Low [11, Theorem 2] and Jafarpour et al. [13]. Here the selected short differences must sum to zero over the reals, rather than merely modulo 2π .

Lemma 4.2 (Real-angle lift criterion). *Suppose (18) holds on every line. There are real angles θ_i representing the phasors and satisfying (13) if and only if*

$$\sum_{\{i,j\} \in C} \varepsilon_{C,ij} k_{ij} = 0 \quad (22)$$

for each fundamental cycle of a spanning forest of G . Equivalently, there exist real vertex variables a_i satisfying

$$a_{\text{root}} = 0 \quad \text{in each component,} \quad a_i - a_j = k_{ij} \quad \text{on each oriented line } j \rightarrow i. \quad (23)$$

Whenever they exist, these root-normalized variables are unique integers.

Proof. For a real lift within the limits, both $\theta_i - \theta_j$ and δ_{ij} belong to $(-\pi, \pi)$ and represent the same argument. They are therefore equal, and their sum around every cycle is zero. Equation (21) proves necessity. Conversely choose one argument at a root of each tree of the forest and extend θ along its edges by $\theta_i - \theta_j = \delta_{ij}$. These angles represent the given phasors. Each non-tree edge and its unique forest path form a fundamental cycle. Its zero sum in (21) forces the same difference equation on that edge. Every line difference is thus its principal difference and satisfies its limit.

For the equivalent formulation, propagate $a_i - a_j = k_{ij}$ along the forest from $a_{\text{root}} = 0$. This uniquely fixes every a_i as a sum of signed integer crossing counts, including $a_i = 0$ at isolated vertices. Each remaining edge equation holds exactly when the crossing sum on its fundamental cycle is zero. Conversely, vertex differences telescope to zero on every cycle. In particular, no integer quantifier is needed: integrality follows from the edge equations and the root normalization. The angles $\theta_i = \alpha_i + 2\pi a_i$ then give an explicit real lift by (20). For an empty graph, all conditions are vacuous. $\square$

Theorem 4.3. *The problem ACPF belongs to $\exists\mathbb{R}$ for rational cosine limits $c_{ij} \in (-1, 1]$. Its existential encoding uses $O(|N| + |E|)$ real variables, atomic predicates, and monomial occurrences, all of degree at most two, and has polynomial total bit length.*

Proof. Use variables e_i, f_i, r_i subject to (17), the injection intervals applied to (15)–(16), and (18). For each oriented line introduce a real k_{ij} and encode (19) by a Boolean combination of polynomial comparisons: either its first condition and $k_{ij} = 1$, or its second condition and $k_{ij} = -1$, or the negations of both conditions and $k_{ij} = 0$. Add the vertex variables and equations (23). Lemmas 4.1–4.2 prove equivalence to real-angle feasibility. There are four real variables per vertex, one per line, one normalization per component, and one difference equation per line. The crossing rule has constant size per line, and the expanded injection expressions have $O(|N| + |E|)$ monomial occurrences in total. Thus the stated linear counts hold, including disconnected and empty graphs. Variable indices and rational input coefficients still contribute to the total bit length; we claim polynomial, not linear, total bit length. Clearing denominators preserves degree at most two and has polynomial bit cost. All quantified variables are real; the crossing rule and root-normalized difference equations enforce the stated discrete values of the crossing variables and vertex shifts. $\square$

The vertex representation is the elementary potential form of a zero cycle-sum condition. It avoids listing long fundamental cycles in the formula; the crossing rule supplies its connection to rational phasor coordinates.

4.3 Zero reactive injection and the hardness transfer

For purely resistive lines with $g_{ij} > 0$ and no shunts, the polar form of (15)–(16) is

$$P_i = \sum_j g_{ij} (v_i^2 - v_i v_j \cos(\theta_i - \theta_j)), \quad (24)$$

$$Q_i = - \sum_j g_{ij} v_i v_j \sin(\theta_i - \theta_j). \quad (25)$$

Lemma 4.4 (Equal angles under a real lift). *In a purely resistive, shunt-free network, suppose $v_i > 0$, $Q_i = 0$ at every bus, and $|\theta_i - \theta_j| < \pi$ on every line. Then angles are constant on each connected component. The active injections are exactly $P_i(v)$ from (1). Conversely, every positive resistive voltage profile extends to an AC profile with these injections and $Q = 0$ by taking all angles zero.*

Proof. Multiply (25) by $-\theta_i$ and sum over buses. Pairing the two contributions of each line yields

$$0 = - \sum_i \theta_i Q_i = \sum_{\{i,j\} \in E} g_{ij} v_i v_j (\theta_i - \theta_j) \sin(\theta_i - \theta_j). \quad (26)$$

Each summand is nonnegative, and $t \sin t > 0$ for $0 < |t| < \pi$. Since $g_{ij} v_i v_j > 0$, every line difference is zero. Substitution in (24) gives (1). The converse follows directly from (24)–(25). $\square$

The sine balance in (25) is also the equilibrium equation of a weighted oscillator network; see Dörfler et al. [10] for this connection and cohesive-angle analysis. The elementary identity (26) supplies the exact statement needed here, with real bus angles and line differences strictly below π . It does not require strictly positive cosine or a stability assumption.

Theorem 4.5. *The problem ACPF is $\exists\mathbb{R}$ -complete. Hardness holds on simple graphs of maximum degree three, with $b_{ij} = h_i = t_i = 0$, $q_i^- = q_i^+ = 0$ at every bus, $c_{ij} = 0$ on every line, and the fixed finite conductance, magnitude, and active-injection data of Theorem 3.1.*

Proof. Take the resistive instance of Theorem 3.1, retain its graph and data, and add the stated AC data. The cosine input zero represents the real-angle limit $\pi/2$. Lemma 4.4 proves equivalence in both directions, and Theorem 4.3 proves membership. $\square$

Thus $\text{ACPF} \in \text{NP}$ would imply $\text{NP} = \exists\mathbb{R}$. This conclusion concerns the specified AC model with variable magnitudes and resistive lines. It does not classify the lossless, fixed-magnitude model, whose nonlinear variables are angles alone.

Corollary 4.6 (One-sided reactive intervals). *Theorem 4.5 remains true if every reactive-injection interval $[0, 0]$ is replaced by the positive-width interval $[0, 1]$. More generally, in a purely resistive, shunt-free network, requiring all reactive injections to be nonnegative, or all to be nonpositive, forces each of them to be zero.*

Proof. Antisymmetry of the sine terms in (25) gives $\sum_i Q_i = 0$. Numbers of one common weak sign with zero sum all vanish. Conversely, zero belongs to $[0, 1]$, so the feasible set is unchanged. $\square$

The same replacement applies to the two restricted AC formulations below. This removes the need to specify singleton reactive intervals, while the network identity still enforces exact zero reactive injections. It is a one-sided saturation argument, not a robustness assertion for symmetric intervals $[-\varepsilon, \varepsilon]$.

4.4 Principal angles: a counterexample and a positive-window result

Define *principal-angle AC feasibility* by (15)–(18) and the injection intervals, without the cycle equations (22). This is a polynomial feasibility problem and hence belongs to $\exists\mathbb{R}$. The distinction from ACPF is material.

Proposition 4.7 (Winding prevents an unrestricted transfer). *On a unit-conductance n -cycle, $n \geq 3$, the phasors $U_k = \exp(2\pi i k/n)$ have unit magnitudes and satisfy*

$$Q_k = 0, \quad P_k = 2(1 - \cos(2\pi/n)) > 0. \quad (27)$$

Their principal line differences have absolute value $2\pi/n$ and winding number one. Consequently, for every fixed rational $c \in (-1, 1)$ there is a rational-data instance with line cosine limit c that is principal-angle AC feasible but whose corresponding resistive instance is infeasible.

Proof. At each bus, the two sine terms in (25) cancel and the cosine terms coincide, giving (27). The oriented sum around the cycle is 2π , so Lemma 4.2 excludes a real lift with these principal differences. Given $c < 1$, choose $n \geq 3$ large enough that $2\pi/n \leq \arccos c$. Put unit magnitude bounds and zero reactive injections at all buses, and choose rational numbers $0 < a < P_k < b$ as active injection bounds $[a, b]$. Such numbers exist by density of $\mathbb{Q}$; the powers in (27) themselves need not be rational. The phasors give principal-angle feasibility, whereas every resistive voltage is one and its injection is zero, outside $[a, b]$. For an explicit instance with singleton rational injections, take $n = 4$, $c = 0$, and $[p_k^-, p_k^+] = [2, 2]$. $\square$

This example disproves an equal-angle implication from fixed positive principal windows; it does not assert a complexity classification for every such fixed-window class. At the other endpoint, the bound strictly below π in Lemma 4.4 is sharp: a single unit-conductance line with unit magnitudes and real angles $0, \pi$ has $Q = 0$ and active injection 2 at both buses.

A positive principal window can nevertheless enforce the needed real lift when its width is allowed to depend on the graph size.

Lemma 4.8 (Small cycle budgets exclude winding). *Suppose principal angle limits $\gamma_{ij} < \pi$ satisfy $\sum_{\{i,j\} \in C} \gamma_{ij} < 2\pi$ for every fundamental cycle C of some spanning forest. Then every phasor profile satisfying those limits has a real lift satisfying the same limits.*

Proof. The principal difference sum on C is a multiple of 2π by (21). Its absolute value is at most the stated sum of limits and therefore strictly less than 2π . It must vanish. Lemma 4.2 applies. In particular, every forest has this property without any additional restriction on its limits below π . $\square$

Corollary 4.9 (Strictly positive principal windows). *Principal-angle AC feasibility is $\exists\mathbb{R}$ -complete even on simple graphs of maximum degree three with purely resistive lines, no shunts, zero reactive injections, the fixed numerical data of Theorem 3.1, and the common cosine limit*

$$c_n = 1 - \frac{1}{n^2}, \quad n = |N| \geq 2. \quad (28)$$

The angle window is strictly positive. Its cosine has $O(\log n)$ bits; unlike the other data, these cosines do not belong to a fixed finite set.

Proof. For $t \in [0, \pi]$, concavity of sine on $[0, \pi/2]$ gives $\sin(t/2) \geq t/\pi$, and therefore $1 - \cos t = 2 \sin^2(t/2) \geq 2t^2/\pi^2$. Writing $\gamma_n = \arccos c_n$, we obtain $0 < \gamma_n \leq \pi/(\sqrt{2}n)$. Every simple cycle has at most n edges, so the sum of its limits is at most $\pi/\sqrt{2} < 2\pi$. Lemma 4.8 supplies a real lift, and Lemma 4.4 proves equivalence to the resistive instance. Take that instance from Theorem 3.1. If needed, add isolated buses with magnitude bounds $[1/2, 2]$ and active injection bounds $[-85, 85]$ to ensure $n \geq 2$. These buses are feasible and use the original finite alphabet. Computing (28) takes polynomial time. Membership follows from the rectangular formulation. $\square$

4.5 A rectangular model with bus-angle boxes

There is also a hardness transfer using only linear angle restrictions in rectangular coordinates. Choose a reference bus in each connected component and require

$$e_i \geq f_i, \quad e_i \geq -f_i \quad (i \in N), \quad f_{\text{ref}} = 0, \quad e_{\text{ref}} > 0. \quad (29)$$

The positive lower magnitude bounds imply $e_i > 0$, so these conditions are exactly the angle box $\theta_i \in [-\pi/4, \pi/4]$ relative to a reference angle zero. Their real line differences lie in $[-\pi/2, \pi/2]$.

Corollary 4.10. *The rectangular AC feasibility problem with (29) is $\exists\mathbb{R}$ -complete under the same graph, admittance, magnitude, and injection restrictions as Theorem 4.5, without separate line-angle constraints. In this restricted class every feasible profile has $f_i = 0$ and $e_i = v_i$ at every bus, and its feasible set is exactly the resistive voltage set in these coordinates.*

Proof. The formulation is polynomial. The box supplies real angles with the required line differences, so Lemma 4.4 makes them constant on each component. Its reference angle makes that constant zero. Conversely every feasible resistive profile with $e_i = v_i$, $f_i = 0$ satisfies the box and all AC constraints. $\square$

The three transfers have different input conventions. Real line limits permit the fixed choice $c_{ij} = 0$ and require the winding equations for membership. Bus-angle boxes give a direct polynomial encoding with fixed coefficients. Positive principal-only windows give a direct polynomial encoding with the size-dependent cosine (28). None of these equivalences assumes that small numerical reactive residuals are exactly zero.

5 Planarity, unit conductance, and long cycles

The arithmetic construction can be made planar without losing its control of the entire solution set. A subsequent electrical subdivision removes the conductance 2 and all short cycles. The restrictions in the next theorem hold simultaneously.

Theorem 5.1. *For every fixed positive integer r , RPF is $\exists\mathbb{R}$ -complete on connected simple planar bipartite graphs of maximum degree three and girth at least r , with every conductance equal to 1. All voltage and injection interval endpoints belong to finite rational sets depending only on r . Moreover, every bounded ETR-INV solution set is rationally equivalent to the feasible voltage set of such a network, with inverse given by projection onto its original variable buses. The network is constructible in polynomial time from the source instance, for fixed r .*

Here an acyclic graph has infinite girth. In particular, the theorem does not assert hardness on trees: its hard graphs can have arbitrarily many cycles. We prove the statement in three steps.

5.1 Planarizing bounded arithmetic while preserving solutions

We use the crossover from the proof of Dobbins et al. [9, Theorem 2.1 and Figure 3]. Its equations are

$$X + Y = Z, \quad X + Y' = Z, \quad X' + Y = Z. \quad (30)$$

They force $X' = X$, $Y' = Y$, and $Z = X + Y$, with unique auxiliary values. Figure 2 gives a planar incidence drawing with the four wire terminals in crossing order. The bounds matter: the cited planar feasibility theorem uses a bounded-witness promise. For preservation of solution sets we instead impose explicit bounds throughout the following construction.

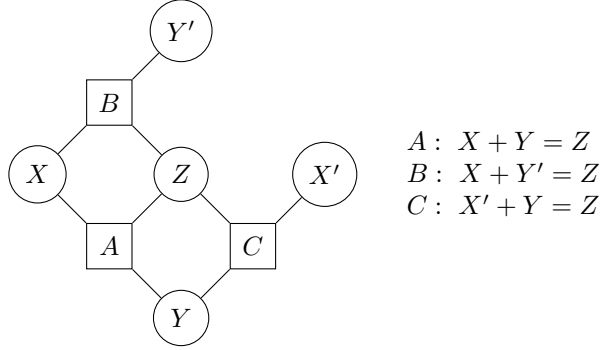


Figure 2: An arithmetic crossover. Circles are variables and squares are equations. The terminal order around the exterior is X, Y, X', Y' . The two transmitted values lie in $[1/2, 2]$; only Z needs $[1, 4]$.

Lemma 5.2. *From a bounded ETR-INV instance one can construct, in polynomial time, an addition-inversion system with planar incidence multigraph and variable intervals in $\{[1/2, 2], [1, 4]\}$, whose solution set is rationally equivalent to the original one by a unique extension and coordinate-projection inverse. All variables in the new system lie in $[1/2, 4]$.*

Proof. Draw the source incidence multigraph with one edge per variable occurrence, including repeated occurrences in an equation. A polygonal drawing in general position with polynomially many crossings can be constructed in polynomial time: start with distinct rational vertex locations and slightly perturb straight edges, separating coincident strands and multiple crossing points. There are only quadratically many pairs of edges. No crossing lies at a vertex. Regard an edge as a wire transmitting its variable value to that occurrence. Replace each crossing by (30) in a disjoint small disk. Each uninterrupted wire segment carries a variable; at an original variable vertex all its initial segments carry that original variable. At an original equation vertex the incident segment values replace the corresponding occurrences. A segment variable can be placed along its corridor, with its incidence edges running within that corridor. Figure 2 shows that all replacements are planar and introduce no further crossings.

Keep every original variable and every transmitted segment value in $[1/2, 2]$. Give each local sum Z the interval $[1, 4]$. The three equations at every crossing force equality of the corresponding incoming and outgoing values; thus projection recovers precisely the original bounded system. Conversely, a source assignment has a unique extension: copy its values along every wire and set each local sum to the sum of its two wire values. These assignments satisfy the indicated intervals. The maps are polynomial with rational coefficients in the forward direction and coordinate projections in the reverse direction, hence continuous. The same reasoning includes empty solution sets and unused original variables. $\square$

5.2 A planar electrical realization

For a system from Lemma 5.2, set $C = 9/2$ and write $\bar{x} = C - x$. This involution preserves $[1/2, 4]$. Use the gadgets of Section 3 with the changes in Table 3. The root bus of each arithmetic variable keeps its actual source interval; every other path value bus uses $[1/2, 4]$. Thus original and wire variables retain their tighter bounds even though their electrical copies use the larger interval.

Bus type	Voltage interval	Fixed injection, if any
Original or wire root	$[1/2, 2]$	none
Crossing-sum root	$[1, 4]$	none
Other path value	$[1/2, 4]$	none
Copy pin and C_I	$[1, 1]$	$2 - C = -5/2$
Addition pin	$[1, 1]$	$3 - C = -3/2$
I	$[1/2, 4]$	-1
W	$[1, 4]$	none
D	$[1, 1]$	$5 - 3C = -17/2$

Table 3: Data for the planar construction. “None” means the same redundant interval $[-85, 85]$ as before. Gadget lines and conductances are unchanged.

The copy equation is $a + b = C$, and the addition equation is $x + y + \bar{z} = C$. At an inversion gadget, C_I gives $v_I = x$, the equation at I gives $v_W = h(x) = 2x - 1 + 1/x$, and the equation at D is

$$4 - v_W - 2(C - x) - (C - y) = 5 - 3C, \quad \text{hence } y = v_W - 2x + 1 = 1/x.$$

If $x, y \in [1/2, 4]$ satisfy $xy = 1$, then in fact both lie in $[1/2, 2]$. Therefore the previously proved bound (10) still guarantees the voltage interval on W in every source solution. This argument is only needed for solutions; the enlarged construction does not claim that $h(x)$ lies in $[1, 4]$ for every $x \in [1/2, 4]$.

To preserve planarity, choose the order of requests on each variable path from its incidence embedding, rather than from the order of equations in an input list. Take disjoint disks about all incidence vertices and narrow disjoint corridors about their edges. Around a variable disk, list the occurrence ports in their cyclic order, breaking that order at any point. Run its alternating copy path just inside the boundary in this linear order; insert up to two path extensions to supply each requested parity. Place each requested value bus at its port. Unrequested intermediate value buses and pinned buses lie between successive ports, and the path has a break between its last and first ports. It therefore encloses no region that would obstruct a corridor. Each requested bus has just one outward gadget line.

An addition disk contains a three-leaf star. An inversion disk contains the tree of Figure 1, with three external leaves: two distinct complemented copies of x and one of y . Replace the x incidence corridor by two adjacent strands. A tree with three specified leaves can be embedded in a disk with those leaves in either cyclic order (reflect an embedding if needed); with three leaves these are all cyclic orders. Thus its required ports can be joined without crossings. If names coincide, keep the occurrences as adjacent distinct strands within their incidence corridors and allocate distinct path buses. The same construction handles a normalized equation $xx = 1$; it never identifies its three external buses.

This disk-and-corridor substitution proves planarity, simplicity, and maximum degree three. Every free injection remains redundant by (11). All fixed-injection equations have the same reversible algebra as before, so extension to network voltages is rational, unique, and continuous, with the

arithmetic root voltages as inverse projection. The finite data sets are explicitly given in Table 3, together with conductances 1, 2 and the free injection interval.

Lemma 5.3. *The networks just constructed can be made connected and planar, preserving their rational solution-set correspondence, degree bound, and finite data.*

Proof. Each nonempty component contains an arithmetic root bus. That root is an endpoint of its copy path and receives at most one gadget line, so its degree is at most two; its injection is free. In each component choose one such root, add a new bus of voltage 1 and free injection, and join it to that root by a unit-conductance line. Join the new buses in a path, also with unit conductance. Every modified degree is at most three. Modified old injections and all new injections satisfy (11) throughout the voltage box; the lines between new buses have zero voltage drop. Thus no constraint on old voltages changes, and the new voltages are uniquely fixed.

For planarity, the chosen root is incident with some face of its component; choose that face as the exterior face before arranging the components in disjoint disks along the new path. Each root-to-new-bus line can leave its component through this face, and the path runs outside the disks. This gives a planar drawing. If the arithmetic system has no variables, its solution set is the single point of $\mathbb{R}^0$; use one isolated voltage-1 bus with injection zero. This also meets all claims, allowing the singleton interval $[0, 0]$ in the fixed injection alphabet. $\square$

5.3 Electrical subdivision

Lemma 5.4. *Fix a positive integer L . In any network above, replace each conductance-1 line by a path of $4L$ unit-conductance lines, and each conductance-2 line by a path of $2L$ unit-conductance lines. Give each new internal bus voltage interval $[1/2, 4]$ and injection zero, and divide every old injection interval by $4L$. The new feasible voltage set is rationally equivalent to the old one by affine extension and coordinate-projection inverse. The new graph is simple, bipartite, and has girth at least $6L$; connectedness, planarity, and the maximum-degree-three bound are preserved.*

Proof. Consider a replacement path of length s with endpoint voltages a, b . Its positive internal voltages and zero injections give $2v_j - v_{j-1} - v_{j+1} = 0$. Hence uniquely

$$v_j = (1 - j/s)a + (j/s)b \quad (0 \leq j \leq s).$$

These convex combinations lie in $[1/2, 4]$. The endpoint current is $(a - b)/s$, equal to the old line current divided by $4L$. Summing at each old bus proves that its new injection is its old injection divided by $4L$. This proves both directions and the affine correspondence.

Subdivision preserves planarity and connectedness and does not increase old degrees. All internal degrees are two. Since every path length is even, color all old vertices alike and alternate colors along each path to obtain a bipartition. Every simple cycle in the new graph traverses full replacement paths and projects to a simple cycle in the old simple graph, which has at least three edges. Each path has at least $2L$ edges, proving the girth bound. For fixed L , the construction has linear size in the old graph and retains a finite data alphabet, obtained by scaling the old injection endpoints and adjoining zero. $\square$

Proof of Theorem 5.1. Apply bounded planarization, the planar electrical realization, and Lemma 5.3. Then apply Lemma 5.4 with $L = \max\{1, \lceil r/6 \rceil\}$. Every step preserves the entire solution set by a rational homeomorphism whose inverse reads earlier coordinates. All steps have polynomial size for fixed r . Hardness follows from ETR-INV, and membership remains Proposition 2.1. $\square$

Corollary 5.5. *The graph and conductance restrictions of Theorem 5.1 also hold simultaneously for the AC hardness results with fixed real-angle limits and for the bus-angle-box model. All numerical data can again be chosen from fixed finite sets. They also hold for the principal-only model of Corollary 4.9, whose positive cosine limit has polynomial bit length and depends on the final number of buses.*

Proof. Apply the corresponding AC transfer to the final subdivided network. For the principal-only transfer, every source with a named variable already produces at least two buses after connection and subdivision, so no isolated padding is needed. For the zero-variable source, instead use two voltage-1 buses with zero active and reactive injections, joined by one unit-conductance line, and set $c_2 = 3/4$. This graph is connected, simple, planar, and bipartite, with degree one and infinite girth; its feasible magnitude set is a singleton. Apart from this exception, the transfers change no graph or conductance and preserve the feasible voltage magnitudes. Zero reactive injections may also be replaced by the one-sided intervals of Corollary 4.6. $\square$

6 Feasible-set universality and exact algebraic voltages

The reduction preserves more than a yes-or-no answer. Its rational solution-set correspondence transfers both topology and fields of definition. We call two real sets *rationally equivalent* if there is a homeomorphism between them whose coordinate functions, and those of its inverse, are rational functions with rational coefficients, with denominators nonzero on their respective domains. A set is *basic closed over $\mathbb{Q}$* if it is described by a finite conjunction of polynomial equalities and weak inequalities with rational coefficients. General semialgebraic descriptions may also use disjunctions and strict inequalities. This distinction matters for rational equivalence.

Lemma 6.1 (Bounded arithmetic for basic closed sets). *Every compact basic closed set $S \subseteq \mathbb{R}^d$ over $\mathbb{Q}$ is rationally equivalent to the solution set of a bounded ETR-INV instance. The equivalence can retain, for every original coordinate t_i , a designated replacement coordinate $x_{j(i)} = 1 + \varepsilon t_i$, where $\varepsilon > 0$ is rational. All added coordinates are uniquely determined rational functions of the original coordinates.*

Appendix A proves the lemma using the conjunction-only arithmetic constructions underlying Lemmas C–G of Abrahamsen and Miltzow [1]. It gives the gate identities, checks their ranges, and avoids Boolean preprocessing. We assert existence here, without a size bound for an arbitrary description of S .

Theorem 6.2. *A compact semialgebraic set defined over $\mathbb{Q}$ is rationally equivalent to the feasible voltage set of a rational RPF instance if and only if it is basic closed over $\mathbb{Q}$. For every prescribed fixed girth bound, the instance can satisfy all the structural and finite-data restrictions of Theorem 5.1.*

Proof. For sufficiency, compose Lemma 6.1 with Proposition 3.3, or with the stronger construction of Theorem 5.1. Composition preserves rationality and continuity in both directions. Necessity follows from Proposition 6.3 below, because every rational RPF voltage set is basic closed over $\mathbb{Q}$. Empty sets are included. $\square$

Proposition 6.3 (Basic closedness is preserved). *If a compact set S is rationally equivalent over $\mathbb{Q}$ to a basic closed set T over $\mathbb{Q}$, then S is basic closed over $\mathbb{Q}$.*

Proof. The empty case is immediate. Otherwise write the equivalence and its inverse as $F : S \rightarrow T$ and $G : T \rightarrow S$. Collect the polynomial denominators of the coordinates of F . For each denominator

of G , substitute F and clear the forward denominators, without cancellation, to obtain a polynomial numerator. Add these numerators to the collection. Every collected polynomial is nonzero on S . Compactness supplies a rational $\eta > 0$ smaller than all their absolute values on S .

Let S' consist of all ambient points t satisfying three conditions: every collected polynomial has square at least η^2 ; $F(t) \in T$; and $G(F(t)) = t$. The first condition makes all expressions in the other two well defined. Clearing their denominators by positive even powers turns them into polynomial equalities and weak inequalities over $\mathbb{Q}$. Hence S' is basic closed. The inverse identities give $S \subseteq S'$. Conversely, if $t \in S'$, then $F(t) \in T$ and $t = G(F(t)) \in S$. Thus $S' = S$. $\square$

The restriction in Theorem 6.2 cannot be dropped. Consider the compact three-quadrant set

$$S = [-1, 1]^2 \cap (\{(x, y) : x \geq 0\} \cup \{(x, y) : y \geq 0\}). \quad (31)$$

It is not even locally basic closed at the origin. To see this, suppose finitely many polynomial weak inequalities describe it near the origin; equalities can be replaced by pairs of inequalities. Every nonzero defining polynomial p with $p(0) = 0$ has a lowest nonzero homogeneous part H that is nonnegative in the three included open quadrants. Its degree cannot be odd: the second and fourth quadrants are negatives of each other, which would force H to vanish on an open set. Its degree is therefore even, so H is nonnegative in the remaining quadrant as well. Choose a direction in the excluded lower-left quadrant avoiding the zero sets of these finitely many nonzero homogeneous polynomials. All their values there are strictly positive. Along a sufficiently short segment of that ray, every p is positive; polynomials with positive constant term also remain positive, and zero polynomials impose no condition. This contradicts exclusion of that segment.

Theorem 6.4 (Topological universality). *Every compact semialgebraic set is semialgebraically homeomorphic to the voltage feasible set of a rational RPF instance satisfying all restrictions of Theorem 5.1, for any prescribed fixed girth bound. The homeomorphism need not be rational.*

Proof. The empty case follows from Theorem 6.2. A nonempty compact semialgebraic set has a semialgebraic triangulation by a finite simplicial complex K ; see Ohmoto and Shiota [19, Theorem 1.1 and Section 1.2]. Let its vertices be $1, \dots, N$. The standard simplex realization of K is

$$T_K = \left\{ z \in \mathbb{R}^N : z_i \geq 0, \sum_i z_i = 1, \prod_{i \in F} z_i = 0 \text{ for every nonface } F \subseteq \{1, \dots, N\} \right\}. \quad (32)$$

The support of z is a face exactly when all the nonface equations hold. Thus T_K is the union of the standard simplices indexed by K . The barycentric map from these standard simplices to any geometric realization of K is a piecewise linear homeomorphism: it is affine and bijective on each simplex, and these maps agree on common faces. Consequently T_K is semialgebraically homeomorphic to the original set. It is compact and basic closed over $\mathbb{Q}$, so Theorem 6.2 applies. No bound on the size of the triangulation or the list of nonfaces is asserted. $\square$

The structured voltage sets therefore need not be connected, contractible, or zero-dimensional. The equal-angle AC transfers preserve these magnitude sets. Fixing a reference angle in each connected component gives the same conclusions for rectangular voltages in the bus-angle-box model. Rational equivalence retains algebraic information that a semialgebraic homeomorphism can change; this is why the two universality statements have different scopes.

Theorem 6.5. *Let α be a real algebraic number. There is a rational RPF instance with a unique feasible voltage vector v such that every v_i belongs to $\mathbb{Q}(\alpha)$ and $\mathbb{Q}(v_j) = \mathbb{Q}(\alpha)$ for some designated*

bus j . Consequently, a uniquely feasible rational network can have a designated bus voltage of any prescribed algebraic degree over $\mathbb{Q}$. All structural restrictions of Theorem 5.1 can be imposed. The same statement holds for unique AC magnitudes, and for unique rectangular voltages in the reference-fixed bus-angle-box model.

Proof. First suppose $\alpha \notin \mathbb{Q}$. Choose a nonzero polynomial in $\mathbb{Q}[t]$ having α as a root and rational endpoints $a < b$ whose interval contains α and no other root. Its root condition together with $a \leq t \leq b$ defines the singleton $\{\alpha\}$. Lemma 6.1 gives a bounded ETR-INV solution set rationally equivalent to this singleton, so it too is a singleton, say $\{x\}$. Rationality of the forward map gives $x_i \in \mathbb{Q}(\alpha)$ for every coordinate.

The designated-coordinate conclusion of the lemma recovers an original coordinate by a rational affine rescaling of one replacement coordinate. Thus $\alpha = Ax_j + B$ for some $A, B \in \mathbb{Q}$. Since α is irrational, $A \neq 0$, and hence $\mathbb{Q}(x_j) = \mathbb{Q}(\alpha)$. The network construction has a unique extension of x , with rational coordinate functions, and it retains x_j as the voltage at its original variable root. The planarization, connectors, and subdivisions retain that root coordinate and add only uniquely determined rational functions of the old coordinates. This proves the claims for irrational α .

For rational α , a single voltage-1, zero-injection bus is sufficient: its unique voltage generates $\mathbb{Q} = \mathbb{Q}(\alpha)$ and its graph meets every stated restriction. To obtain degree $d \geq 2$, take the positive real root of $t^d - 2$; this polynomial is irreducible by Eisenstein's criterion at 2. Degree one is the preceding rational example. Finally, the exact AC transfers preserve precisely the RPF magnitude set, and reference-fixed angle boxes force every phase to zero. $\square$

The designated-coordinate assertion uses the affine recovery property, not merely preservation of the field generated by all coordinates. The theorem is an existence statement and asserts no additional size bound on the affine recovery coefficients. It also does not by itself exclude polynomially verifiable symbolic certificates: the relevant obstruction to such certificates is the complexity consequence in Corollary 3.4.

7 Residuals, approximate certificates, and reactive stability

Exact feasibility and small residuals answer different questions. We make their relationship explicit, first for the arithmetic reduction and then for general bounded resistive networks. Throughout this section the voltage bounds remain exact, including singleton bounds at pinned buses. Only injection constraints are relaxed.

For a nonempty rational voltage box $B = \prod_i [\ell_i, u_i]$, define

$$\rho_G(v) = \max_i \text{dist}(\mathcal{P}_i(v), [p_i^-, p_i^+]), \quad \gamma_G = \min_{v \in B} \rho_G(v). \quad (33)$$

Continuity and compactness imply that the minimum is attained and that $\gamma_G = 0$ exactly when the network is feasible. For a normalized source instance Φ , define

$$R_\Phi(x) = \max\{|x + y - z| : x + y = z \text{ in } \Phi; |xy - 1| : xy = 1 \text{ in } \Phi\}, \quad \gamma_\Phi = \min_{x \in [1/2, 2]^n} R_\Phi(x). \quad (34)$$

A maximum over no constraints is zero, including for an empty network.

7.1 A quantitative correspondence for the actual gadgets

Theorem 7.1. *For the ordinary construction of Section 3, with m normalized equations,*

$$\frac{\gamma_\Phi}{10(6m + 1)} \leq \gamma_G \leq 2\gamma_\Phi. \quad (35)$$

Proof. Given any source-box point x , extend it using exact complementary copies, $v_I = x$, and $v_W = h(x) = 2x - 1 + 1/x$, whether or not it solves Φ . The voltage bounds hold by (10). The free-injection bounds are redundant throughout the box. All copy, C_I , and I equations hold exactly. An addition bus has residual $|x + y - z|$, and an inversion bus D has residual $|y - 1/x| \leq 2|xy - 1|$. Taking a minimizing source point gives the upper bound.

Conversely, let a network-box point have residual ε and read each source variable at its root. A copy pin has voltage exactly 1, so its injection residual bounds the error in its neighboring complement sum by ε . There are at most $6m$ path extensions in the whole network. Induction along a path therefore shows that each requested copy differs from its intended root value or complement by at most $\delta = 6m\varepsilon$. The addition equation now gives source residual at most $\varepsilon + 3\delta$.

In an inversion gadget, write x, y for the two root values. The residual at C_I gives $|v_I - x| \leq \varepsilon + \delta$. Dividing the residual at I by $v_I \geq 1/2$ gives

$$|v_W - h(v_I)| \leq 2\varepsilon.$$

On $[1/2, 2]$, $|h'| \leq 2$, so $|v_W - h(x)| \leq 2\varepsilon + 2(\varepsilon + \delta)$. The residual at D , including its x copy of weight two and its y copy of weight one, gives

$$|y - (v_W - 2x + 1)| \leq \varepsilon + 3\delta.$$

Combining these inequalities and $x \leq 2$ yields

$$|xy - 1| \leq 2(\varepsilon + 3\delta + 2\varepsilon + 2(\varepsilon + \delta)) = 10\varepsilon + 10\delta.$$

This also bounds every addition residual. Apply the resulting inequality $\gamma_\Phi \leq 10(6m + 1)\rho_G(v)$ to a minimizing network-box point. If $m = 0$, both residual minima are zero and the conclusion still holds. $\square$

The theorem bounds the loss in transferring a residual; it does not supply a lower bound on γ_Φ for arbitrary infeasible formulas. In fact, even the fixed-data networks of Section 3 can have extremely small positive residual minima.

7.2 Infeasible fixed-data networks with doubly exponential accuracy

Theorem 7.2. *For every integer $k \geq 0$ there is an infeasible connected network G_k with the fixed numerical data and degree bound of Theorem 3.1, at most $68k + 67$ buses and $72k + 72$ lines, and a rational voltage vector in its exact voltage box for which every injection constraint except one holds exactly and*

$$0 < \gamma_{G_k} \leq \rho_{G_k}(v) < 2^{-2^k}. \quad (36)$$

The strict lower bound concerns the minimum; the exhibited residual is only an upper bound on that minimum.

Proof. Use the variables $t, h, x_0, \dots, x_k$ and a_j, b_j, c_j for $0 \leq j < k$, all in $[1/2, 2]$. Impose

$$\begin{aligned} tt &= 1, & h + h &= t, & t + h &= x_0, \\ a_j + h &= x_j, & x_j b_j &= 1, & a_j + b_j &= c_j, & x_{j+1} + h &= c_j \quad (0 \leq j < k), \\ x_k t &= 1. \end{aligned} \quad (37)$$

The first three equations force $t = 1$, $h = 1/2$, $x_0 = 3/2$. Each group of four then forces

$$x_{j+1} = x_j + 1/x_j - 1.$$

Writing $x_j = 1 + \delta_j$ gives $\delta_0 = 1/2$ and $\delta_{j+1} = \delta_j^2/(1 + \delta_j) > 0$. The last equation requires $x_k = 1$, a contradiction.

Nevertheless, assign precisely these recurrence values and set $a_j = x_j - 1/2$, $b_j = 1/x_j$, and $c_j = x_j + 1/x_j - 1/2$. Then

$$1 < x_j \leq 3/2, \quad 1/2 < a_j \leq 1, \quad 2/3 \leq b_j < 1, \quad 3/2 < c_j \leq 5/3,$$

so every voltage-source bound holds. All source equations except the last hold exactly. More explicitly, $\delta_j = 1/d_j$, where $d_0 = 2$ and $d_{j+1} = d_j(d_j + 1)$. Induction gives $d_k \geq 2^{2^k}$.

Apply the ordinary electrical construction and its canonical extension to this nonsolution. The proof of Theorem 7.1 shows that only the final inversion bus D violates its injection constraint, by exactly

$$1 - 1/x_k = \frac{1}{d_k + 1} < 2^{-2^k}.$$

All voltage intervals hold exactly. The source system is infeasible, hence so is the network, and compactness of its voltage box makes its residual minimum strictly positive. There are $4k + 3$ variables and $4k + 4$ equations; the size bounds follow from Theorem 3.1. The source incidence graph is connected, and each variable path and equation gadget is connected with all incidence attachments present. Thus the network is connected as well. $\square$

Its standard graph encoding has length $O((k + 1) \log(k + 2))$. Therefore no bound of the form $2^{-\text{poly}(s)}$, with s the input length, can be a universal lower bound on positive residual minima in this fixed-data class. Ruling out these infeasible instances solely by a universal residual threshold can require exponentially many accuracy bits in the number of buses. This is not a lower bound on exact decision algorithms: the displayed family has the short algebraic infeasibility proof just given.

There is also a general lower bound at the doubly exponential scale. We record it to identify the scale of the preceding construction, using a general theorem about polynomial minima.

Proposition 7.3. *For a rational network with $n \geq 1$ buses, put $q = n + 1$. There is an explicitly constructible integer coefficient bound H with polynomial bit length in the input such that, with $M = 4n + 2$, $\tilde{H} = \max\{H, 2q + 2M\}$, and $\tau = \lceil \log_2 \tilde{H} \rceil$, infeasibility implies*

$$\gamma_G \geq 2^{-q^{4^q}(\tau+q+4)}. \quad (38)$$

For fixed-data bounded-degree networks this yields $\log_2(1/\gamma_G) \leq 2^{O(n)}$ whenever $0 < \gamma_G < 1$. Together with Theorem 7.2, the worst-case scale of this logarithm in that class is $2^{\Theta(n)}$.

Proof. Let $U = \max_i u_i$, $D = \max_i \sum_j g_{ij}$, and $A = \max_i \max\{|p_i^-|, |p_i^+|\}$. On the positive voltage box, $|\mathcal{P}_i(v)| \leq U^2 D$, so $B_0 = 1 + U^2 D + A$ bounds ρ_G from above. Consider the compact set

$$T = \{(v, t) : v \in B, 0 \leq t \leq B_0, t \geq p_i^- - \mathcal{P}_i(v), t \geq \mathcal{P}_i(v) - p_i^+ \ (i \in N)\}.$$

It is connected: each point joins vertically to (v, B_0) , and the whole top face $B \times \{B_0\}$ lies in T and is convex. The minimum of the polynomial t on T is γ_G . There are $M = 4n + 2$ weak polynomial inequalities, of degree at most two. Clear denominators in each by a positive integer, and let $H \geq 1$ bound the absolute values of the resulting integer coefficients, also including the objective t . Rational arithmetic gives $\log H$ polynomial in the input length.

The polynomial-minimum bound of Jeronimo et al. [15, Theorem 1.1], applied in dimension $q \geq 2$ with even degree bound $d = 2$, gives for a nonzero minimum

$$\gamma_G \geq (2^{4-q/2} \tilde{H} 2^q)^{-q^{4^q}}.$$

The logarithm of the base is at most $\tau + q + 4$, proving (38). With a fixed rational data alphabet and bounded degree, B_0 and the cleared coefficient bound can be chosen constant. Then $\tau = O(\log(n+2))$, and the upper estimate on $\log_2(1/\gamma_G)$ follows. For the lower estimate, the family above has linearly many buses in k (in particular its distinct variable roots number $4k + 3$), and its logarithmic residual exceeds 2^k . $\square$

7.3 Short rational certificates for a gap promise

Proposition 7.4. *Let $\varepsilon > 0$ be rational. Every exactly feasible rational RPF instance has a rational voltage vector in its exact box with $\rho_G(v) \leq \varepsilon/2$, of bit length polynomial in the network input and the encoding length of ε . Consequently, the promise problem whose yes cases have $\gamma_G = 0$ and whose no cases have $\gamma_G > \varepsilon$ has polynomial-size certificates verifiable in deterministic polynomial time. No answer is prescribed when $0 < \gamma_G \leq \varepsilon$.*

Proof. An empty network is an immediate yes instance and needs no voltage coordinates. Otherwise put $U = \max_i u_i$ and $D = \max_i \sum_j g_{ij}$. For v, w in the voltage box, differentiating (1) gives

$$\left| \frac{\partial \mathcal{P}_i}{\partial v_i} \right| \leq 3UD, \quad \sum_{j \neq i} \left| \frac{\partial \mathcal{P}_i}{\partial v_j} \right| \leq UD.$$

The segment between v and w remains in the box, so

$$|\mathcal{P}_i(v) - \mathcal{P}_i(w)| \leq 4UD \|v - w\|_\infty. \quad (39)$$

The distance to an interval is 1-Lipschitz, and the same bound holds for the change of ρ_G . Choose a dyadic mesh $\eta = 2^{-b}$ with $\eta \leq \varepsilon/(2 \max\{1, 4UD\})$ and $b \geq 0$. Round each coordinate of an exactly feasible point down to this mesh and then clamp it to its own rational interval. The resulting point stays in the box, moves by at most η , and has residual at most $\varepsilon/2$. Singleton bounds are preserved by clamping. The exponent b and all resulting rational-coordinate lengths are polynomial in the input encoding and in that of ε .

A verifier checks the exact rational voltage bounds and evaluates all injections with rational arithmetic, accepting when the residual is at most $\varepsilon/2$. The stated bit bound gives completeness on promised yes instances. No certificate can be accepted when $\gamma_G > \varepsilon$, giving soundness on promised no instances. $\square$

This is a certificate-existence result, not an algorithm for finding such a point. It does not put exact RPF in NP, nor does it decide all instances of an unpromised approximate feasibility problem at a sharp tolerance. In particular, it does not resolve the approximate-membership question of Bienstock and Verma [6, Section 1.3]: their question relaxes sine equations in a lossless, fixed-magnitude model and does not impose the promise used here.

7.4 Stability of the equal-angle transfer

The exact reactive argument also has a quantitative version. It applies to real bus angles with short edge differences; a principal-angle winding solution does not satisfy that hypothesis.

Proposition 7.5. *Consider a connected purely resistive AC network with no shunts and $n \geq 2$ buses. Let $0 < \ell \leq v_i \leq U$, and suppose the real angle differences on all lines lie in $[-\pi/2, \pi/2]$. Center*

the angles so that $\sum_i \theta_i = 0$, let q be their reactive-injection vector, and let $\lambda_2 > 0$ be the smallest positive eigenvalue of the conductance Laplacian L_g . Then

$$\|\theta\|_2 \leq \frac{\pi \|q\|_2}{2\ell^2 \lambda_2}, \quad (40)$$

$$0 \leq P_i(v, \theta) - \mathcal{P}_i(v) \leq \frac{\pi^2 U^2 \|q\|_2^2}{8\ell^4 \lambda_2} \quad (i \in N). \quad (41)$$

One may replace the constants $\pi/2$ and $\pi^2/8$ by the rational constants 2 and 2, respectively. Moreover,

$$\lambda_2 \geq \frac{2g_{\min}}{(n-1)^2}, \quad g_{\min} = \min_{\{i,j\} \in E} g_{ij}. \quad (42)$$

Thus entirely rational bounds can be computed directly from the graph data. For disconnected networks these statements apply componentwise. An isolated bus has $q_i = 0$ and zero active discrepancy.

Proof. Put $s = \theta^T L_g \theta = \sum_{\{i,j\}} g_{ij} (\theta_i - \theta_j)^2$. The energy identity (26) and $t \sin t \geq (2/\pi)t^2$ for $|t| \leq \pi/2$ give

$$(2/\pi)\ell^2 s \leq -\theta^T q \leq \|\theta\|_2 \|q\|_2 \leq \sqrt{s/\lambda_2} \|q\|_2.$$

If $s = 0$, the centered angles vanish and both conclusions are immediate. Otherwise divide by $\sqrt{s}$ to obtain $s \leq \pi^2 \|q\|_2^2 / (4\ell^4 \lambda_2)$, which also gives (40). From (24) and $0 \leq 1 - \cos t \leq t^2/2$,

$$0 \leq P_i(v, \theta) - \mathcal{P}_i(v) = \sum_j g_{ij} v_i v_j (1 - \cos(\theta_i - \theta_j)) \leq (U^2/2)s.$$

This proves (41). Using the weaker inequality $t \sin t \geq t^2/2$ gives the stated rational constants.

Finally, along a simple path of at most $n-1$ edges, Cauchy–Schwarz gives $(\theta_i - \theta_j)^2 \leq (n-1)s/g_{\min}$. For centered angles,

$$\|\theta\|_2^2 = \frac{1}{n} \sum_{i < j} (\theta_i - \theta_j)^2 \leq \frac{(n-1)^2}{2g_{\min}} s.$$

Minimizing the Rayleigh quotient over nonzero centered vectors proves (42). Components can be centered independently because the electrical quantities depend only on edge differences. $\square$

Corollary 7.6. *Consider an AC instance obtained from the ordinary m -equation resistive construction, with $n \geq 1$ buses, real line differences in $[-\pi/2, \pi/2]$, and exact voltage bounds. If a state has active-injection interval violation at most ε and $\|q\|_\infty \leq \eta$, its root voltage assignment has source residual at most*

$$10(6m+1)(\varepsilon + 256n(n-1)^2\eta^2). \quad (43)$$

Proof. Here $\ell = 1/2$, $U = 4$, and $g_{\min} \geq 1$ in every nonsingleton component. The rational bounds of Proposition 7.5, and $\|q\|_2^2 \leq n\eta^2$, bound the active discrepancy by $256n(n-1)^2\eta^2$, also for smaller components. Subtracting this discrepancy from the actual AC injections bounds $\rho_G(v)$ by the parenthesis in (43). The pointwise reverse estimate proved in Theorem 7.1 gives the result. Singleton components have zero discrepancy, including the case $n = 1$. $\square$

These estimates quantify the equal-angle argument by elementary energy and Poincaré inequalities. They control the error induced by a small reactive residual, but establish neither hardness with symmetric positive reactive tolerances nor the absence of the winding solutions in Proposition 4.7.

8 Conclusion

Bounded addition and inversion can be implemented by a sparse resistive network while retaining every source solution and every coordinate needed to recover it. The resulting exact feasibility problem is $\exists\mathbb{R}$ -complete even under the simultaneous structural restrictions of Theorem 5.1. The model allows independently prescribed voltage and injection bounds, including simultaneous singletons. Within this model, the graph and coefficient restrictions do not suffice to make exact feasibility easier than general existential-real decision. The solution-preserving electrical realization is the main contribution; the arithmetic and topological tools on which it builds are established.

The same correspondence explains the algebraic and numerical behavior of feasible voltage sets. Rational equivalence realizes precisely the compact basic closed sets over $\mathbb{Q}$, while semialgebraic homeomorphism realizes all compact semialgebraic topologies. Unique voltages can have arbitrarily high algebraic degree. Separately, the ordinary bounded-degree, fixed-data construction gives infeasible networks with doubly exponentially small injection residuals. This matches the general separation scale for that class; no residual-preserving planarization is asserted. Exact decision, tolerance-based numerical evidence, and certificates for a gap promise consequently require different guarantees.

For AC models, the transfer depends on the meaning of an angle limit. An explicit rational quadratic formula encodes the established zero-winding condition using linearly many variables and predicates. Reactive balance then gives both an exact equal-angle theorem and a quantitative stability estimate. Principal-angle constraints alone allow winding states, whereas the stated bus-angle boxes and size-dependent principal windows support separate hardness transfers. These conclusions classify exact feasibility for the specified models and delimit residual-based certification; they do not predict typical solver performance or cover other operating models that exclude the prescribed voltage and injection combinations used by the reduction.

A A verified bounded arithmetic construction

This appendix proves Lemma 6.1. The arithmetic identities follow the conjunction-only reductions of Abrahamsen and Miltzow [1, Lemmas C–G and Figures 2–3]; we give the details needed for rational equivalence and use compactness and continuity for the range choices. In particular, we do not use its Boolean normal-form step or quantitative range lemma.

A.1 Unique polynomial evaluation and scaling

Clear positive rational denominators in a basic closed description of S . Evaluate its integer polynomials by a finite arithmetic circuit with equations of the forms

$$z = 1, \quad x + y = z, \quad xy = z,$$

and require the appropriate output coordinates to be zero or nonnegative. The constant zero is fixed by $0 + 1 = 1$, and a negative value is introduced by $x + (-x) = 0$. Integer constants are formed by repeated addition and negation. A zero output can be imposed by $z + 0 = 0$. Each circuit coordinate is a uniquely determined polynomial in the original coordinates. The solution set of this conjunctive system is therefore the graph of a polynomial map on S , with coordinate projection as inverse. It is compact. Write $M \geq 1$ for a bound on the absolute values of all circuit coordinates on that set; the empty case can be represented directly by the inconsistent bounded equation $x + x = x$.

Choose a sufficiently small dyadic number $\delta = 2^{-\ell} > 0$, as specified below, and a dyadic $\varepsilon = \delta 2^{-k}$ with $k \geq 0$ and $\varepsilon M \leq \delta$. Replace each circuit variable x by $w_x = \varepsilon x$. Introduce a variable $d = \delta$,

a zero variable via $0 + d = d$, and the halving chain from d to ε . These constants are all uniquely determined. Replace the circuit equations by

$$\begin{aligned} x = 1 : \quad w_x + 0 &= \varepsilon, \\ x + y = z : \quad w_x + w_y &= w_z, \\ xy = z : \quad w_x w_y &= q, \quad \varepsilon w_z = q. \end{aligned}$$

Here q is a new variable for that multiplication. Nonnegativity becomes $w_x \geq 0$. Every solution of this new system recovers a solution of the old one by $x = w_x/\varepsilon$; the converse extension is unique and has $q = \varepsilon^2 z$. Thus all its variables lie in $[-\delta, \delta]$ when $\delta \leq 1$. This is a consequence of the equations and the compact original set, not an additional bound that could remove solutions. We need only existence of ε , and make no complexity claim about the length of its halving chain.

A.2 Shifted addition and multiplication

All variables in the final construction have the common interval $[1/2, 2]$. A constant c in the following displays denotes a dedicated variable fixed to that value by the construction. First fix the variable c_1 by $c_1 c_1 = 1$, an inversion equation of ETR-INV; positivity excludes the root -1 . We then denote this fixed variable by 1. The additions

$$\frac{1}{2} + \frac{1}{2} = 1, \quad 1 + \frac{1}{2} = \frac{3}{2}, \quad \frac{3}{4} + \frac{3}{4} = \frac{3}{2}, \quad 1 + 1 = 2$$

fix the other displayed constants. Fix $2/3$ by the inversion $(3/2)(2/3) = 1$.

For each small variable s , retain $A_s = 1 + s$ and introduce

$$A_s + \frac{1}{2} = C_s, \quad B_s + \frac{3}{4} = C_s. \tag{44}$$

These uniquely enforce $C_s = s + 3/2$ and $B_s = s + 3/4$. An addition $s + t = u$ is replaced by $B_s + B_t = C_u$. For a product $st = u$, introduce the following equations, with fresh m, f, g, h :

$$\begin{aligned} A_s A_t &= m, & m + \frac{1}{2} &= f, & g + B_t &= f, \\ g + \frac{3}{4} &= h, & B_u + B_s &= h. \end{aligned} \tag{45}$$

Successive substitution gives

$$m = 1 + s + t + st, \quad f = \frac{3}{2} + s + t + st, \quad g = \frac{3}{4} + s + st, \quad h = \frac{3}{2} + s + st.$$

The last equation is precisely $u = st$. To impose $s \geq 0$, add $J_s + 1/2 = A_s$; its built-in bound $J_s \geq 1/2$ is exactly the desired inequality.

It remains to impose $d = \delta$. Construct $D_i = 1 - 2^{-i}$ for $i = 1, \dots, \ell$: start with $D_1 = 1/2$, and use

$$D_i + 1 = E_i, \quad D_{i+1} + D_{i+1} = E_i.$$

All these constants lie in $[1/2, 2]$. With $\Delta = D_\ell = 1 - \delta$, the equation $A_d + \Delta = 2$ fixes $d = \delta$ exactly. The small system's other equations, including its halving chain, are replaced by the addition and product rules above. For δ sufficiently small, all nonconstant coordinates in these rules are in $(1/2, 2)$, except $J_s = 1/2 + s$, whose range $[1/2, 1/2 + \delta]$ follows from $s \geq 0$. The only multiplications still present have their two operands near 1. Their replacement is described next.

A.3 Products from squares, and squares from reciprocals

In the next two displays, each assignment introduces a fresh variable, except that the final output is identified with the desired existing output coordinate. An assignment $v = u - w$ means the addition equation $v + w = u$, and $v = u/2$ means $v + v = u$. Thus subtraction and halving introduce no new allowed operation. The order of the assignments makes all auxiliary values unique.

For inputs a, b near 1, the following construction uses only additions and three squarings:

$$\begin{aligned}
a' &= a + \frac{1}{2}, & b' &= b + \frac{1}{2}, & h_a &= a'/2, & h_b &= b'/2, \\
t &= h_a + h_b, & r &= t - \frac{1}{2}, & s &= r^2, & u &= s + \frac{1}{2}, \\
a_2 &= a^2, & b_2 &= b^2, & a_3 &= a_2 + \frac{1}{2}, & b_3 &= b_2 + \frac{1}{2}, \\
a_4 &= a_3/2, & b_4 &= b_3/2, & c &= a_4 + b_4, & d &= c/2, \\
e &= u - d, & f &= e + e, & o &= f - \frac{1}{2}.
\end{aligned} \tag{46}$$

Indeed $r = (a + b)/2$, $d = (a^2 + b^2 + 1)/4$, and

$$o = 2 \left(\frac{(a + b)^2}{4} + \frac{1}{2} - \frac{a^2 + b^2 + 1}{4} \right) - \frac{1}{2} = ab.$$

At $a = b = 1$, every variable in this display is one of $3/4, 1, 3/2$. In particular all lie strictly inside the allowed interval, and each of the three squared inputs r, a, b is near 1.

For an input a near 1, replace its squaring by

$$\begin{aligned}
b &= a + \frac{1}{2}, & b' &= 1/b, & a' &= 1/a, \\
c &= a' + \frac{1}{2}, & d &= c - \frac{2}{3}, & e &= d + \frac{1}{2}, \\
f &= e - b', & g &= f + f, & h &= g - \frac{2}{3}, \\
i &= 1/h, & j &= i - \frac{1}{2}, & k &= j + \frac{3}{4}, \\
l &= b/2, & o &= k - l.
\end{aligned} \tag{47}$$

Each reciprocal assignment $v = 1/u$ is the allowed equation $uv = 1$. The central identity is

$$h = 2 \left(\frac{1}{a} + \frac{1}{3} - \frac{1}{a + 1/2} \right) - \frac{2}{3} = \frac{1}{a(a + 1/2)}.$$

Consequently $i = a^2 + a/2$ and $o = i - 1/2 + 3/4 - (a + 1/2)/2 = a^2$. At $a = 1$, the variables belong to

$$\{2/3, 3/4, 5/6, 1, 4/3, 3/2, 7/4\} \subset (1/2, 2).$$

All denominators are nonzero there.

For completeness, the range choices can be made before scaling the original circuit. The finitely many rational functions in (47) are continuous with nonzero denominators at $a = 1$, so they all remain in $(1/2, 2)$ on one neighborhood of 1. The finitely many polynomial functions in (46) are continuous at $(1, 1)$, and its three squared inputs approach 1 there. Shrink the product-input neighborhood so that all its variables stay in $(1/2, 2)$ and its squared inputs enter the reciprocal gate's neighborhood. Finally choose a dyadic $\delta > 0$ small enough for the shifted-product inputs A_s, A_t and all the shifted auxiliary coordinates above. This choice is uniform over the finite gate types; it does not depend on the number of circuit gates. The constants constructed by the midpoint and halving chains already have their stated exact ranges and need no continuity argument.

Every remaining constraint is now an addition or inversion with variables in $[1/2, 2]$. On each original solution the formulas give one valid extension. Conversely, every bounded solution forces the

constants, reciprocal gates, square gates, and shifted equations by the identities above. It therefore recovers the small system, and then the original circuit and point of S . Each added coordinate is a rational function of the original point with nonzero denominators on S ; recovery is $t_i = (A_{w_{t_i}} - 1)/\varepsilon$. Both maps are continuous, and this proves Lemma 6.1, including its designated-coordinate claim.

A.4 Why the input must be conjunctive

The general rational-equivalence assertion in Abrahamsen and Miltzow [1, Theorem 1] is broader than the part used here. Its Lemma A replaces weak inequalities by equations with nonnegative auxiliary variables inside their Boolean branches. For example, applying that rule and eliminating a disjunction by a product turns $x \geq 0$ or $y \geq 0$ into

$$(x - u)(y - v) = 0, \quad u \geq 0, \quad v \geq 0.$$

At $(x, y) = (-1/2, 1/2)$, every $u \geq 0$ with $v = 1/2$ satisfies these conditions. Thus the auxiliaries are not uniquely determined; adding the box $[-1, 1]^2$ to the original variables does not repair that failure. The displayed replacement preserves existence of a solution, but it does not give the claimed rational bijection. The basic-closedness obstruction proved in Section 6 shows that unrestricted rational equivalence cannot be restored by choosing a different Boolean preprocessing rule. Our polynomial evaluation starts from a conjunction and defines every auxiliary globally, so it has no inactive branches.

B Reproducible exact checks and illustrative instances

The supplementary directory `checks/` contains four deterministic Python programs using only the standard library. They evaluate rational quantities with exact arithmetic; none invokes an optimization solver. The proofs establish the quantified statements. The programs provide independent finite checks of the actual gadget equations, data restrictions, sign conventions, and boundary cases.

Resistive construction. `check_resistive_exact.py` constructs the original network and computes each bus injection by summing its incident line currents. It checks 12,751 profiles, including 606 source solutions, all variable-identification patterns in one addition or inversion equation, both inversion endpoints, unused variables, an empty instance, long copy paths, high fan-out, and 300 reproducibly generated mixed systems. It verifies distinct occurrence buses, simplicity, maximum degree, fixed numerical data, the size bound, dissipation, and the source-to-network residual identities. Unsatisfying source assignments are included; evaluating such an assignment is not an infeasibility decision procedure.

AC signs and winding. `check_ac_exact.py` checks 2,112 scaled short-arc pairs, including 768 whose absolute principal angle exceeds $\pi/2$, and 14,784 rational-cosine tests. An independent segment-ray intersection calculation verifies line crossings. For 177,168 cycles, a rotated branch cut verifies the winding count; 59,640 cycles have nonzero winding. A further 648 complex-power calculations check the rectangular signs against complex multiplication. Regressions cover axis endpoints, unequal radii, antipodal exclusion, the rational four-cycle counterexample, cancellation of total reactive injection, and the size-dependent positive cosine inputs. For the vertex formulation of zero winding, the suite compares exact propagation of root-normalized shifts against an independent enumeration of all simple-cycle sums in 4,166 labeled graph and crossing-count cases through four vertices. Exactly 1,090 admit shifts; all recovered shifts are integral. These cases include empty and

disconnected graphs and isolated vertices. Further phasor regressions reject the winding four-cycle and recover nonzero integer shifts across the branch cut on a zero-winding cycle. These tests cover the full short-arc convention used here, including obtuse arcs.

Structural and quantitative developments. `check_developments_exact.py` checks 49 generalized crossover profiles, eight generalized inversion profiles including repeated names, connection of three components, and three even-subdivision scales on a weighted triangle. It evaluates 360 perturbed original-network profiles, the infeasible recurrence family for $k = 0, \dots, 10$, and 100 rational Poincaré and voltage-Lipschitz samples. The recurrence implementation has $49 + 43k$ buses and $54 + 47k$ lines; direct injection aggregation gives the stated residual $1/(d_k + 1)$. Its positive residual is evaluated exactly, while infeasibility is established by Theorem 7.2. These checks do not replace the planar embedding proof or the analytic spectral estimates.

Bounded arithmetic and topology. `check_arithmetic_exact.py` checks 1,681 composed multiplication and reciprocal profiles. A complete arithmetic encoding of a disk has 332 variables and 330 final addition or inversion equations. The program checks 49 feasible profiles, 72 outside points, and 49 altered inconsistent profiles against those final equations. It also checks nine dyadic constant-chain configurations and 35 standard-simplex support profiles. These tests cover concrete instances of Appendix A; neither universality nor the triangulation theorem is inferred from sampling.

B.1 Illustrative exact feasible and infeasible instances

Table 4 gives eight small bounded arithmetic systems. Every variable is in $[1/2, 2]$. The six feasible systems have unique source solutions and therefore unique resistive voltage extensions. The two irrational examples illustrate why the proof, rather than a rounded voltage vector, establishes exact feasibility.

Source equations	Exact outcome
$x^2 = 1$	$x = 1$
$x + x = y, xy = 1$	$x = 1/\sqrt{2}, y = \sqrt{2}$
$x + y = z, xy = 1, z^2 = 1$	Infeasible: $z = 1$, but $x + y \geq 2$
$x + x = y, y + y = z, z^2 = 1$	Infeasible: $x = 1/4 < 1/2$
$x + x = y, y^2 = 1$	$x = 1/2, y = 1$
$x + x = y, y + y = z$	$x = 1/2, y = 1, z = 2$
$x + y = z, xz = 1, y^2 = 1$	$y = 1, x = (\sqrt{5} - 1)/2,$ $z = (\sqrt{5} + 1)/2$
$u + u = w, w^2 = 1, uy_i = 1$ for $i = 1, 2, 3$	$u = 1/2, w = 1, y_1 = y_2 = y_3 = 2$

Table 4: Exact source solutions and infeasibility obstructions for small instances. All equalities and interval checks follow in the source domain.

The first, fifth, and eighth rows use positivity to resolve a square equal to one. In the second row, substitution gives $2x^2 = 1$. The third row uses $x + y \geq 2\sqrt{xy} = 2$. The fourth gives $z = 4x = 1$. For the sixth, $z = 4x \leq 2$ and $x \geq 1/2$ force the endpoint solution. The seventh gives $y = 1$ and $x(x + 1) = 1$, with a unique positive root. The displayed feasible values all lie in the source interval, so Lemma 3.2 establishes their exact network outcomes.

Corollary 4.10 gives the same exact outcomes for the corresponding reference-fixed AC instances with bus-angle boxes: every feasible rectangular profile has $f_i = 0$ and $e_i = v_i$. These conclusions

follow from the exact solution correspondence, with no numerical tolerance. The supplementary README gives portable build and checking commands.

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